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Improving Decision Making: Using Databases to Analyze Sales Trends

Scenario:

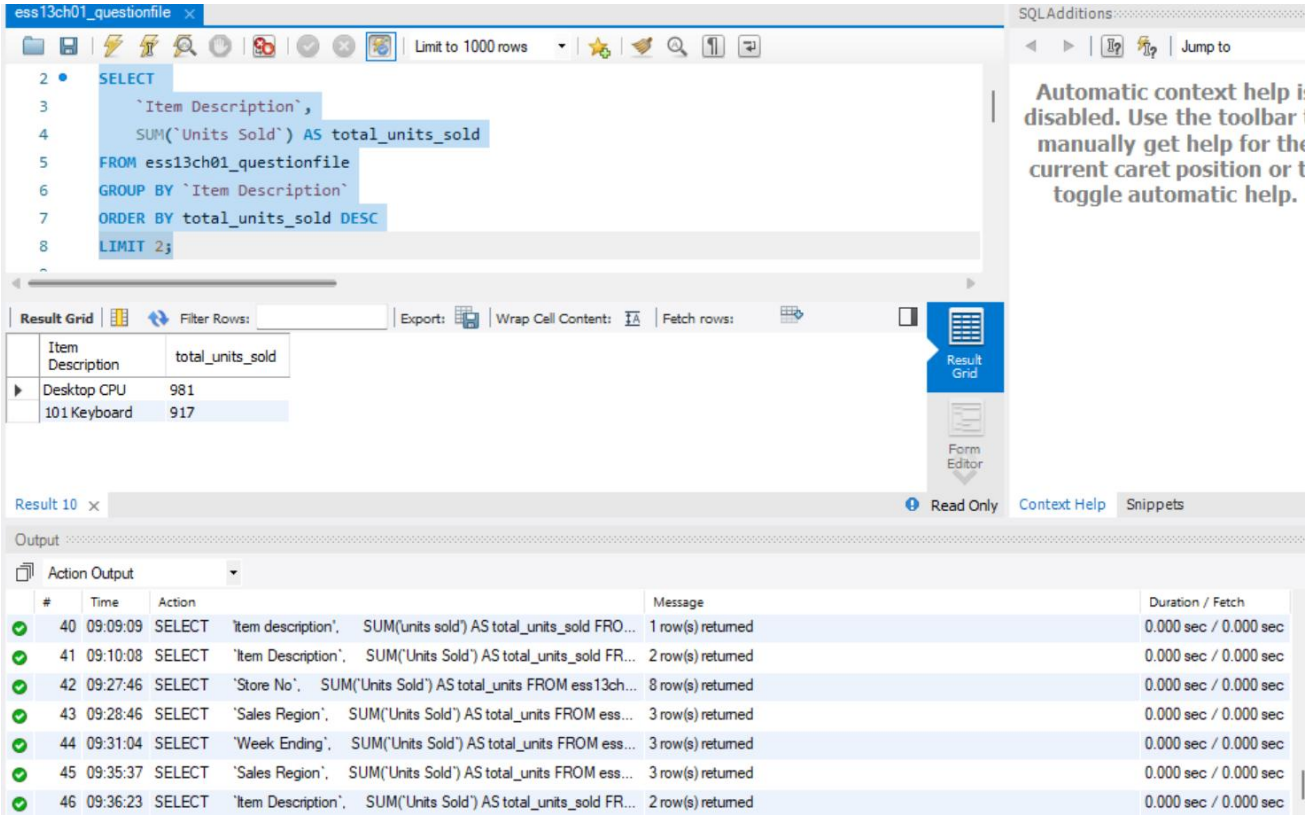
In this project, you will start out with raw transactional sales data and use any tools to develop queries and reports that help managers make better decisions about product pricing, sales promotions, and inventory replenishment. The database file name is Ess13ch01_questionfile.xlsx. The database contains raw data on weekly store sales of computer equipment in various sales regions. The database includes fields for store identification number, sales region, item number, item description, unit price, units sold, and the weekly sales period when the sales were made. Use any programming language to develop some reports and queries to make this information more useful for running the business.

Sales and production managers want answers to the following questions:

1) List the first two products should be restocked. Explain.

Query:

```
SELECT
   `Item Description`,
   SUM(`Units Sold`) AS total_units_sold
FROM ess13ch01_questionfile
GROUP BY `Item Description`
ORDER BY total_units_sold DESC
LIMIT 2;
```



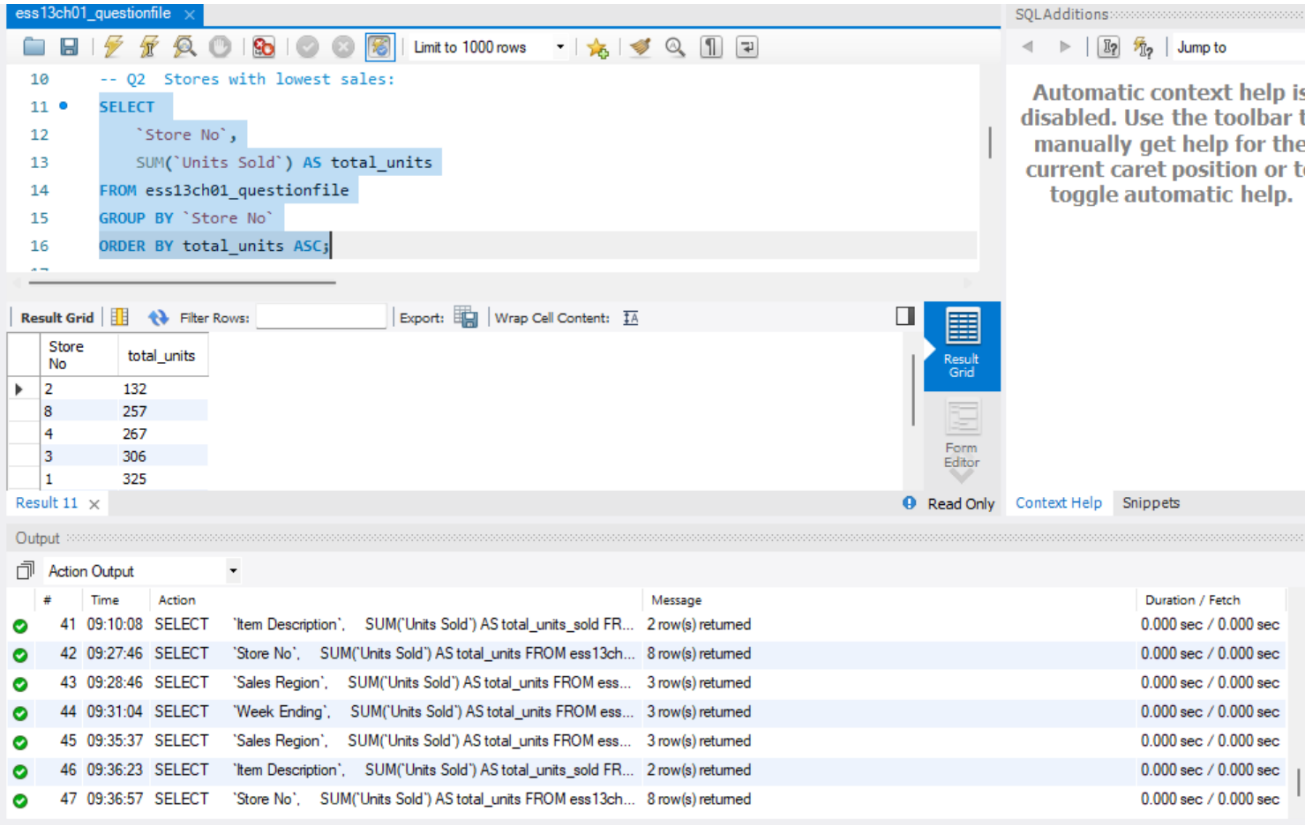
Explanation: Products with the highest overall sales volume should be prioritized for restocking first because they represent the strongest demand.

2) Which stores, and sales regions would benefit from a promotional campaign and additional marketing?

   Query (Stores with lowest sales):

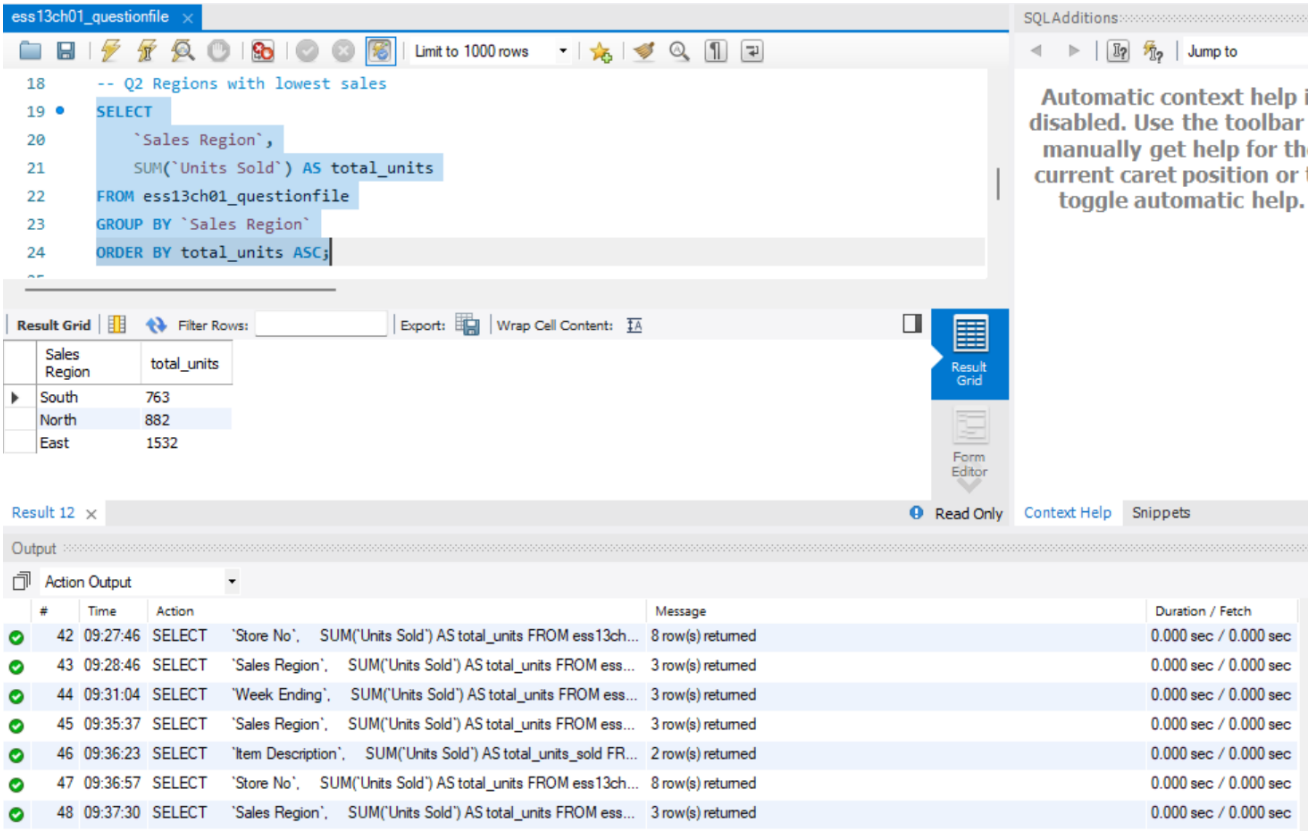
```
SELECT
```

```
    `Store No`,
    SUM(`Units Sold`) AS total_units
FROM ess13ch01_questionfile
GROUP BY `Store No`
ORDER BY total_units ASC;
```



Query (Regions with lowest sales):

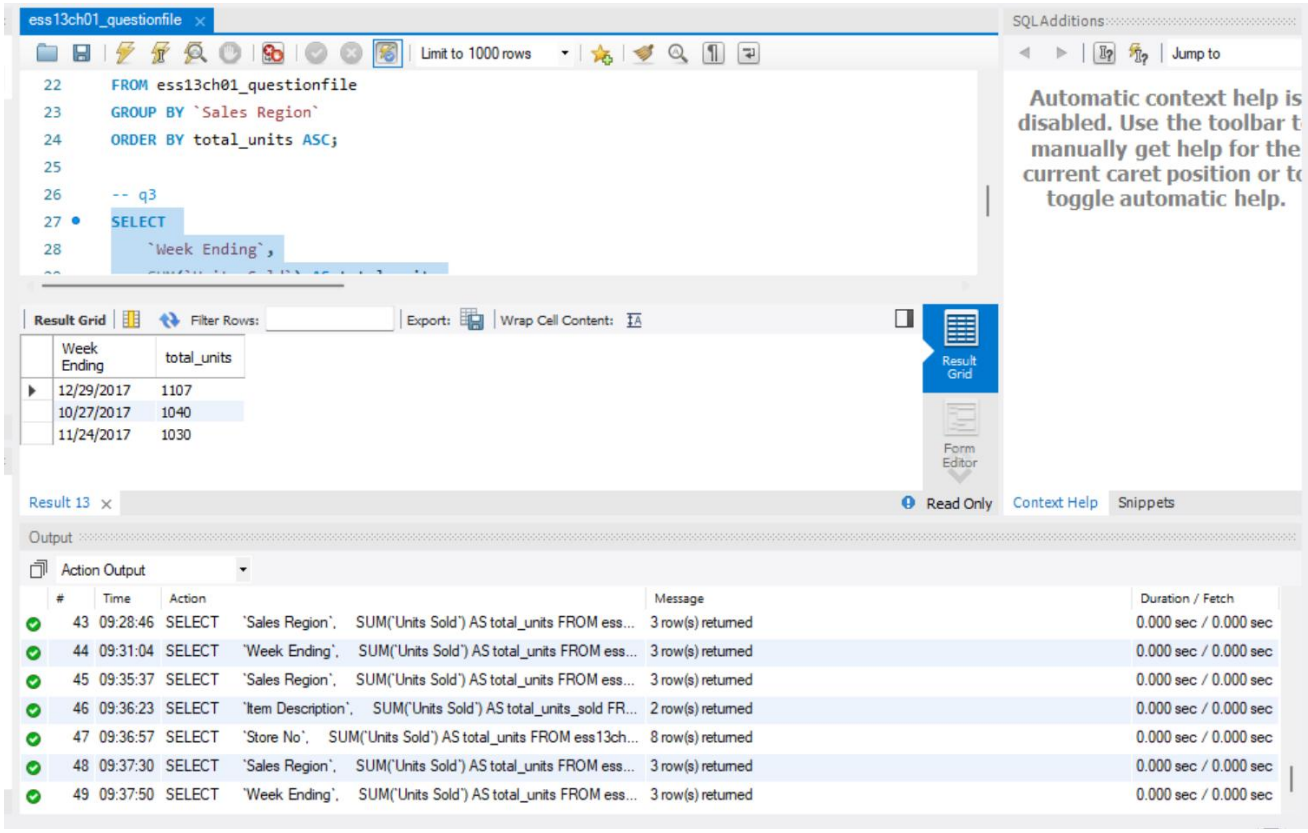
```
SELECT
    `Sales Region`,
    SUM(`Units Sold`) AS total_units
FROM ess13ch01_questionfile
GROUP BY `Sales Region`
ORDER BY total_units ASC;
```



Explanation: Stores and regions with the lowest sales volumes would benefit from promotional campaigns and additional marketing efforts to increase demand.

3) When (what time of year) should products be offered at full price, and when should discounts be used?

```
SELECT
   `Week Ending`,
   SUM(`Units Sold`) AS total_units
FROM ess13ch01_questionfile
GROUP BY `Week Ending`
ORDER BY total_units DESC;
```



Explanation: Weeks with high sales indicate strong demand and allow full-price selling. Weeks with lower sales suggest the need for discounts or promotional pricing.